

# MedGemma-27B vs Bio-Medical-Llama-3-8B — the abnormal-or-normal call

We labeled Harvard’s clinical EEG reports, 101,614 across three hospitals, with MedGemma-27B (prompt v5g, Q4) and compared the result against Harvard’s own Bio-Medical-Llama-3-8B labels for the same reports. There is no human ground truth here, so this is a similarity comparison between two independent language models, not an accuracy one, and a test of how well a model built on our own reports lines up with a different model on a hospital system it has never seen.

This report looks at one finding only: whether the EEG is abnormal or normal. We keep to that finding because it maps cleanly between the two schemas. The slowing findings do not: our focal and generalized non-epileptiform categories bundle slowing together with attenuation, asymmetry, disorganization, encephalopathy, and excessive beta, whereas the Harvard schema files each of those under its own separate column. So a direct focal-slowing or generalized-slowing comparison would be measuring two different things, and we leave it out here (the check behind that is in analysis/mapping\_check.py). The abnormal-or-normal call has no such problem.

We report two numbers: the match rate, the share of reports where the two models make the same abnormal/normal call, and Cohen’s  $\kappa$ , which discounts the agreement you would expect by chance.

## How much the two agree

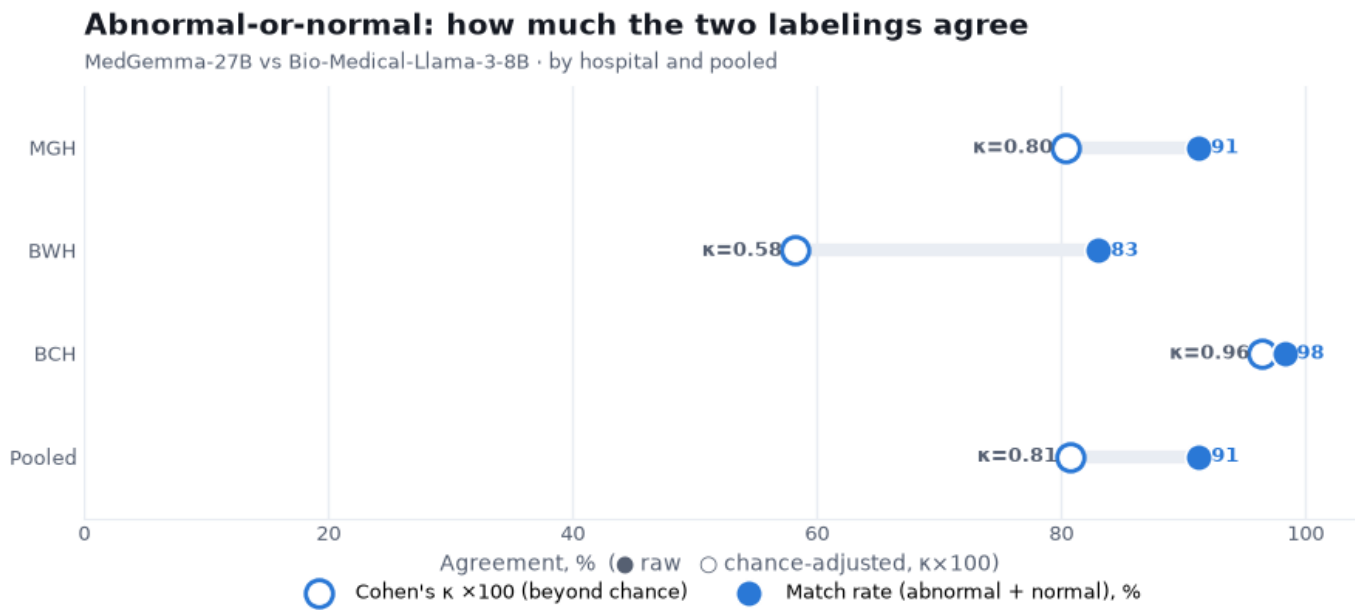


Figure 1: Abnormality agreement by hospital

| Hospital      | Reports        | Match rate | Cohen’s $\kappa$ | MedGemma abnormal | Bio-Medical-Llama abnormal |
|---------------|----------------|------------|------------------|-------------------|----------------------------|
| MGH           | 55,557         | 91%        | 0.80             | 67%               | 66%                        |
| BWH           | 21,035         | 83%        | 0.58             | 78%               | 67%                        |
| BCH           | 25,022         | 98%        | 0.96             | 57%               | 58%                        |
| <b>Pooled</b> | <b>101,614</b> | <b>91%</b> | <b>0.81</b>      | <b>67%</b>        | <b>64%</b>                 |

Pooled across the three hospitals the two models make the same call on **91% of reports, with  $\kappa = 0.81$**  — strong agreement well beyond chance. Two models built by different groups, on reports neither was tuned for, land on the same abnormal-or-normal verdict nine times in ten. The pediatric hospital,

BCH, is almost a lock-step match (98%,  $\kappa = 0.96$ ). BWH is the loosest ( $\kappa = 0.58$ ): there MedGemma calls abnormal noticeably more often than Bio-Medical-Llama (78% vs 67%), so the two part ways on more reports.

Where the reports land

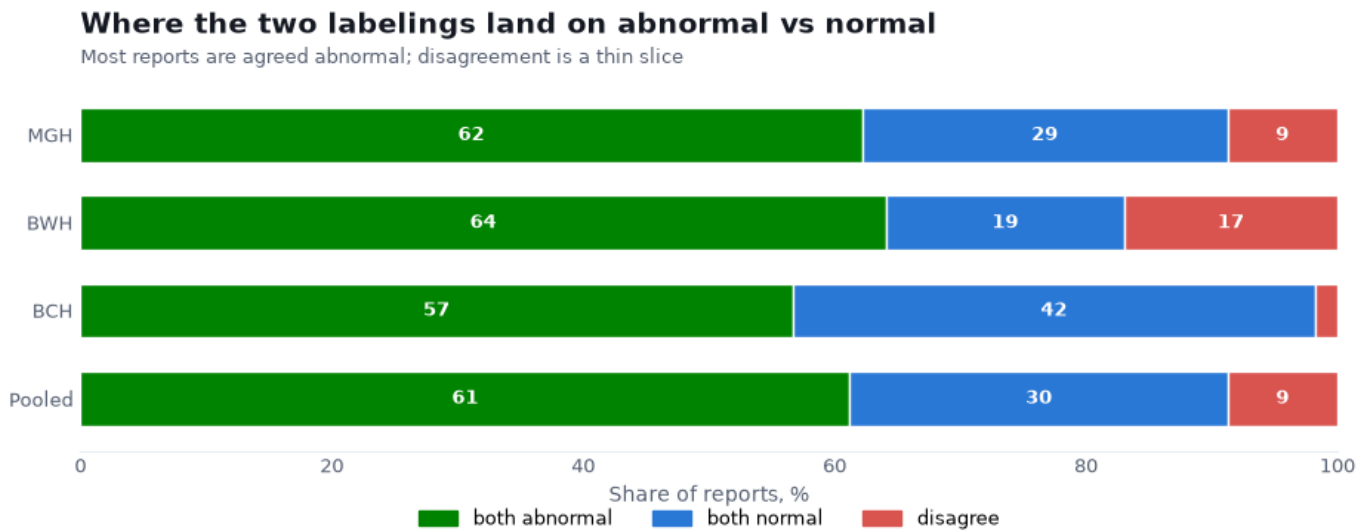


Figure 2: Where they land

Splitting every report three ways: pooled, **61% are agreed abnormal, 30% agreed normal, and only 9% are a disagreement**. Most EEGs in this corpus are abnormal and both models see that; the genuine split is a thin slice, widest at BWH (17%) and almost nothing at BCH (about 1%).

How often each model calls abnormal

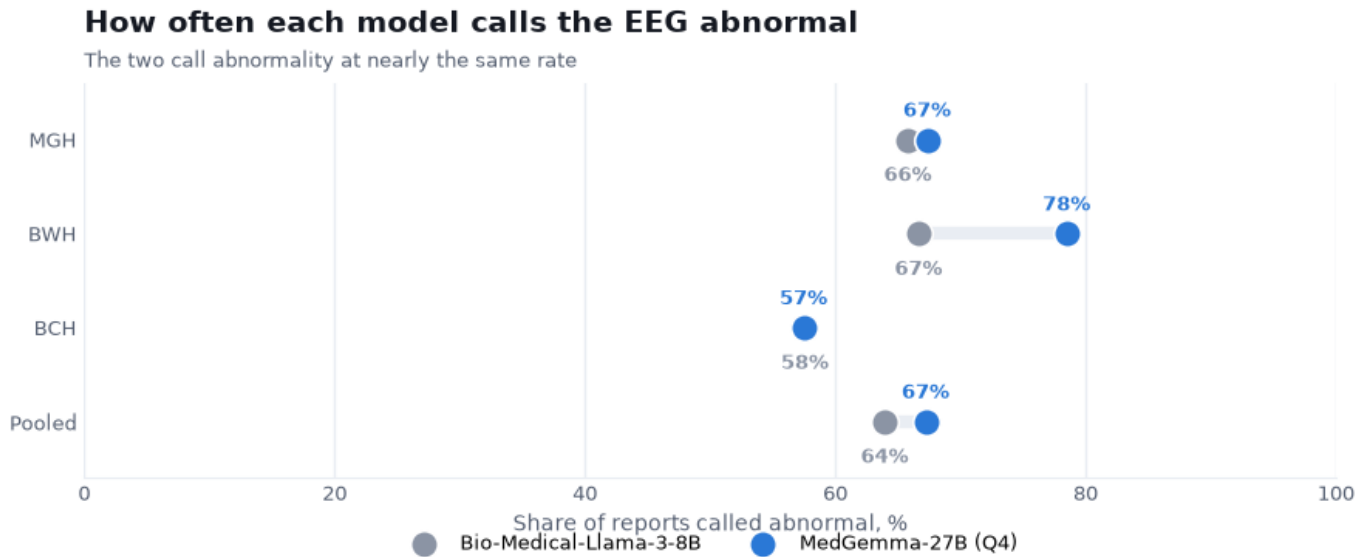


Figure 3: Call rate

The two mark abnormality at nearly the same rate overall (MedGemma 67%, Bio-Medical-Llama 64%), and very close at MGH and BCH. BWH is the one place they diverge, with MedGemma reading more of

those reports as abnormal. That gap is what pulls BWH's  $\kappa$  down while the raw match rate still looks high.

## What this means

The overall abnormal-or-normal call is the finding that carries between the two labelings.  $\kappa = 0.81$  pooled, and 0.96 at BCH, is good evidence that this part of the labeling generalizes past the reports it was built on: an unseen hospital system, a different model, and the two still agree on the headline call. The one soft spot is BWH, where MedGemma leans abnormal more than Bio-Medical-Llama does; since there is no ground truth, we cannot say which is closer to right there.

## Caveats

- Similarity, not accuracy. There is no human label for these reports, so where the two differ neither can be called correct.
- One finding only. The slowing findings are left out because the two schemas define non-epileptiform activity differently, not because the models were checked and disagreed.
- BIDMC (Beth Israel) is excluded — its report text is not in the shared data.

## Reproduce

```
python -m analysis.harvard_abnormality      # the table above and the three figures
python -m analysis.mapping_check             # why the slowing findings are left out
```

MedGemma labels: results/harvard/labels/\*/ (produced by slurm/label\_harvard.sbatch via slurm/submit\_harvard.sh, reading the report text from the HEEDB zips through analysis/harvard\_index.py). Bio-Medical-Llama labels: the \*\_EEG\_with\_reports tables in HarvardEEG.db.

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Pipeline, prompts, and the scripts behind this report: <https://github.com/vbronetskyi/eeg-report-classification>